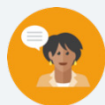


5.9 Congruence in Two-Dimensional Figures – Part 1

Lesson Introduction

 Benchmark	MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.		 Learning Target	I can solve mathematical problems involving congruence.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- How can I use my knowledge of triangle congruence to solve mathematical problems? - How can corresponding parts of congruent polygons be used to determine measures?			
 Lesson Narrative	In this lesson, students will apply their knowledge of triangle congruence and corresponding parts of congruent triangles from the prior lessons in this unit to solve mathematical problems. Students will work together to learn how the use of CPCTC can help to determine measures of a given triangle and extend that into understanding corresponding parts of congruent polygons are congruent. They will then use this knowledge to solve mathematical problems.			
 Component Summary	5.9.1 Warm-Up (~5 min.)	Complete a two-column proof on triangle congruence (<i>SE p. 272</i>)	5.9.3 Guided Instruction (10-15 min.)	Apply corresponding parts of congruent polygons to find unknown angles and lengths (<i>SE p. 273</i>)
	5.9.2 Collaboration (10 min.)	Collaborate on using CPCTC to determine unknown measures (<i>SE p. 273</i>)	5.9.4 Guided Practice (10-15 min.)	Use corresponding parts to solve mathematical problems (<i>SE p. 275</i>)
	5.9.5 Wrap-Up (~5 min.)	Error analysis activity on congruent parts to solve a mathematical problem (<i>SE p. 276</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> Corresponding Parts of Congruent Polygons are Congruent <u>Additional Terms:</u> N/A		(Optional) Markers (Optional) Patty paper	Some of these problems have areas where minor missteps can throw off the entire process. Work through each of these problems on your own, prior to teaching, to look for these errors that students may make.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with decomposing the figures. - Students may not understand which parts in congruent polygons are corresponding parts. - Students may not recall how to use the Pythagorean Theorem.	This lesson is the first of a two-part series. Each problem will require students to use given information to make markings as well as additional information, such as the Pythagorean Theorem.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 5.9.2 Collaboration is a collaborative activity for students to work together on using CPCTC to determine measures. In this activity, students will have the opportunity to think about the given triangle and then pair up with a partner to discuss their work and come to a collaborative solution.	MA.K12.MTR.4.1: Discussion and Reflection During the Wrap-Up activity, students will complete an error analysis activity on congruent parts to solve a mathematical problem.
 Differentiate Instruction	Consider partner pairings by performance on the lessons prior to the proof sections. Placing students together who struggle with the same concept can provide you with an opportunity to work with smaller groups on similar issues. Consider using manipulatives or other tools to deconstruct certain polygons to help students visualize the shapes that comprise them. 5.9.4 Guided Practice has two problems. Consider having struggling students only work on question 1 that has been edited with additional scaffolding included. Samples of scaffolding questions are: - What type of triangle is $\triangle YHD$? - What formula can be used to find a missing side of a right triangle? $\angle HDT$ has the same measure as which angle? - What is the sum of the measures of the three angles of a triangle?	
Lesson Notes:		

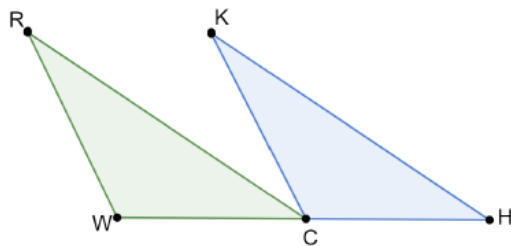
5.9.1 Warm-Up: Bell Work

Component Overview: Students will fill in the missing statements and reasons in a two-column proof on triangle congruence. For this activity, allow students to work independently to determine what they recall from the prior lesson.



5.9.1 Warm-Up: Bell Work

Triangles RCW and KHC are shown.



Given: $RW \parallel KC$, C is the midpoint of $\overline{WH}$, $\overline{WR} \cong \overline{CK}$
 Prove: $\angle WRC \cong \angle CKH$

- Complete the missing statements and reasons in the two-column proof.

Statement	Reason
1. C is the midpoint of $\overline{WH}$	1. Given
2. $\overline{WC} \cong \overline{CH}$	2. Definition of midpoint
3. $RW \parallel KC$	3. Given
4. $\angle RWC \cong \angle KCH$	4. Corresponding Angles Theorem
5. $\overline{WR} \cong \overline{CK}$	5. Given
6. $\triangle RWC \cong \triangle KCH$	6. SAS Congruence Postulate
7. $\angle WRC \cong \angle CKH$	7. CPCTC



For this activity, allow students to try this problem completely on their own to see what they have retained from the previous lesson. Use the information from the formative assessment to determine possible misconceptions your students have.



Consider adjusting this Warm-Up based on student performance in the prior lesson. If students are struggling with the content, consider providing a word bank for them to use to match to the reason only. If students are doing well, consider adding the challenge of having them complete the whole proof without prompts.

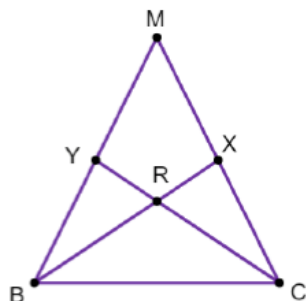
5.9.2 Collaboration: Using CPCTC to Determine Measures

Component Overview: Students will work together to use CPCTC to determine unknown measures. For this activity, students should work with a partner or small group to move through discovery of the content.



5.9.2 Collaboration: Using CPCTC to Determine Measures

Triangle MBC is shown where
 $MX = 3.6$, $MB = 7.2$, $RX = 2$,
 $BC = 6.5$, $RB = 3.9$, $m\angle YBR = 29^\circ$,
 $m\angle CXR = 83^\circ$, and $m\angle MCB = 63^\circ$.



Work with your partner to complete the following.

- How many different triangles are in $\triangle MBC$?
7 Triangles
- If $\triangle BRY \cong \triangle CRX$, then determine the measure of each.
 $RC = \underline{3.9}$ $m\angle RYB = \underline{83^\circ}$
- If $\triangle BXM \cong \triangle CYM$, then determine the measure of each.
 $CY = \underline{5.9}$ $CM = \underline{7.2}$ $m\angle MCY = \underline{29^\circ}$



Visual learners may benefit from using colored markers or highlighters to help visualize different angles and side lengths when doing their markings within the triangle.



When reviewing the answers with students, allow them to share out their thinking to each question. For question 1, consider having each student share one of the 8 triangles that they found.

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons

Component Overview: Students will receive guided instruction on corresponding parts of congruent polygons.



5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons

Any polygon that can be mapped to another polygon using rigid transformations shows that all of the parts of the polygons are congruent. This understanding can then be applied to find measurements of the two congruent polygons.

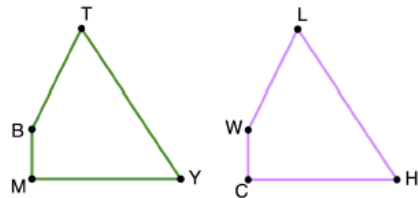


Corresponding Parts of Congruent Polygons are Congruent

If two polygons are congruent, then all corresponding parts of the polygons are congruent.

1. Congruent quadrilaterals $TYMB$ and $LHCW$ are shown.

Some measurements of the angles in quadrilaterals $TYMB$ and $LHCW$ are given.



$m\angle YMB = 90^\circ$,
 $m\angle WLH = 60^\circ$, $m\angle MBT + m\angle CHL = 210^\circ$, $m\angle BTY = (2x - 4)^\circ$,
 $m\angle MYT = (x + 25)^\circ$

- a. Determine each.

$m\angle BTY$

60°

$m\angle CWL$

153°



Students should be familiar with the corresponding parts concepts from prior lessons with triangles. Emphasize this concept when introducing CPCTC and the connection to the definition of congruence, which states that if figures are congruent, then they have equal measures. Reminding students of rigid motions on the coordinate plane may also help students recall that congruent figures would have corresponding parts with equal measures.



What markings did you make in the quadrilaterals with the given information?

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons (continued)

Some measurements of the sides of quadrilaterals $TYMB$ and $LHCW$ are given.

$HC = 5k$, $LW = 4k - 0.5$, $YT = 6k$, and $BM = k + 1$.

- b. Determine the value of k if the perimeter of quadrilateral $TYMB$ is 24.5.

$$k + 1 + 4k - 0.5 + 6k + 5k = 24.5$$

$$k = 1.5$$

- c. What is the length of BT ?

$$BT = 4k - 0.5$$

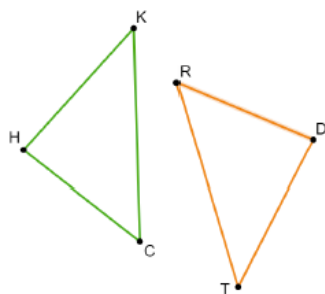
$$BT = 5.5$$

2. The diagram shows $\triangle HKC$ and $\triangle TDR$ where $\triangle HKC \cong \triangle TDR$. Some measurements of the sides are given.

$$KH = 5y - 2,$$

$$DR = 6y - 5, \text{ and}$$

$$TD = 4y + 3$$



Arthur's Work

$$5y - 2 = 6y - 5$$

$$-2 = y - 5$$

$$3 = y$$

$$KH = 5(3) - 2$$

$$KH = 13$$

Arthur used the information to determine the length of KH . His work is shown.

- a. Find Arthur's error.

$\overline{KH} \cong \overline{TD}$ because corresponding parts of congruent polygons are congruent. Arthur used the wrong side when setting up his equation. He used side length RD instead of TD .



Students should be familiar with perimeter, but still consider asking students what that word means. This can benefit ELL students or students who are unfamiliar with the concept. Monitor students' work on solving the equation to determine if remediation is needed for the algebra or if there is a misunderstanding of the geometry in the question.



For question 2, consider allowing students to try this one on their own first after introducing the context. If needed, assist students with marking the triangles using the given information and then releasing them. When reviewing, have students share their rationale and leverage an agree/disagree protocol.

5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons (continued)

- b. Help Arthur understand and learn from his mistake.
Write a brief note to Arthur explaining what his mistake was and how he can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.



Provide sentence stems, such as:

Arthur made a mistake when he _____ because _____.

Arthur should have _____ instead of _____.

The next time Arthur sees _____, Arthur should _____.

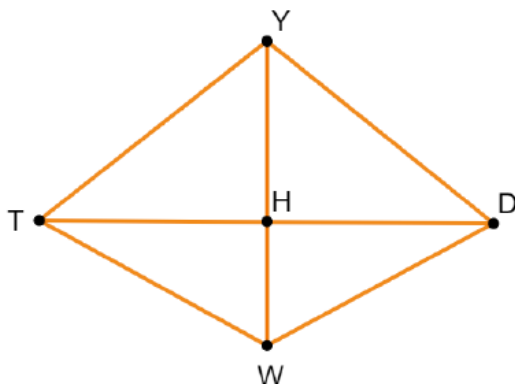
5.9.4 Guided Practice: Corresponding Parts

Component Overview: Students will engage in structured practice on corresponding parts to solve mathematical problems.



5.9.4 Guided Practice: Corresponding Parts

1. Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



- a. How many different triangles are in Quadrilateral $YDWT$?

8 Triangles

- b. If $\triangle DHY \cong \triangle THY$, $YD = \sqrt{130}$, $YH = 7$, and $m\angle HTY = 38^\circ$, determine the following measures.

$$HD = \underline{9} \quad m\angle DYH = \underline{52^\circ} \quad TH = \underline{9}$$

- c. If $\triangle DHW \cong \triangle THW$, $WH = 5$, and $m\angle HWD = 61^\circ$, use these and all other known measurements to determine the following measures.

$$m\angle TWH = \underline{61^\circ} \quad m\angle HTW = \underline{29^\circ} \quad TW = \underline{\sqrt{106}}$$



Consider having struggling students only work on question 1 that has been edited to include additional scaffolding.

Samples of scaffolding questions are,

- What type of triangle is $\triangle YHD$?
- What formula can be used to find a missing side of a right triangle?
- $\angle HDT$ has the same measure as which angle?
- What is the sum of the measures of the three angles of a triangle?

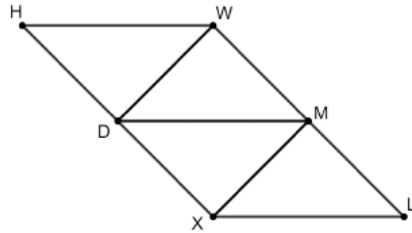


In question 1c, students may think that there is not enough information to find TW . Remind them that they can use the Pythagorean Theorem to find the length of HD , which can then be used to find the length of DW , which will be congruent to TW based on all the information given.

5.9.4 Guided Practice: Corresponding Parts (continued)

Component Overview: Students will complete an error analysis activity on congruent parts to solve a mathematical problem. This activity is designed to be done independently to get student data on their level of comprehension of the content of this lesson.

2. Quadrilateral $HWLX$ is shown.



$HWM \cong LXM$ and $\triangle XMD \cong \triangle WDM$. The perimeter of $HWMX$ is 40.5.

- a. Use the given measurements to find the value of x .

$$HW = 2x - 2, XD = \frac{3}{2}x - 1, HX = 2x + 5, WD = 2x - 2$$

$$2x - 2 + \frac{3}{2}x - 1 + 2x + 5 + 2x - 2 = 40.5$$

$$x = 5.4$$

- b. Find the length of MX .

$$MX = WD = 2x - 2$$

$$MX = 8.8$$



Students may struggle to visualize the image provided in question 2. Encourage students to first draw the two quadrilaterals separately and/or to mark the congruent pieces from the congruency statements.



For an additional challenge, consider having students find the length of additional sides, including the full perimeter of the quadrilateral.

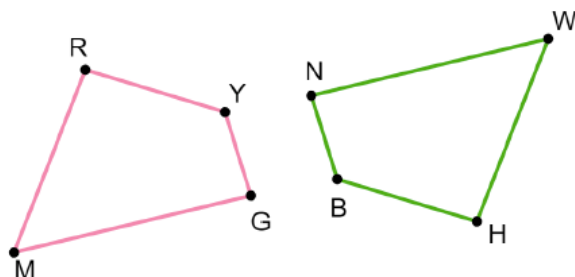
5.9.5 Wrap-Up: Error Analysis

Component Overview: Students will complete an error analysis on congruent parts to solve a mathematical problem. This is designed to be done independently to get student data on their level of comprehension of the content of this lesson.



5.9.5 Wrap-Up: Error Analysis

1. Quadrilaterals $RYGM$ and $HBNW$ are shown.



$RYGM \cong HBNW$ and the following information is also true.

$$m\angle RYG = (10x - 6)^\circ, MG = 2k + 3, NW = 4k - 29, \\ m\angle MGY = (7x - 5)^\circ, m\angle BNW = 86^\circ, HB = k + 9$$

Hector used the given information to write the two equations shown.

$$10x - 6 = 86$$

$$4k - 29 = 2k + 3$$

One of Hector's equations is NOT correct. Determine which one, and correct his error.

The first equation is incorrect. The equation should be $7x - 5 = 86$



This question provides an error analysis on the content from this lesson. The data from this activity can provide information on whether or not a student understands, and to what degree, the content from this lesson.